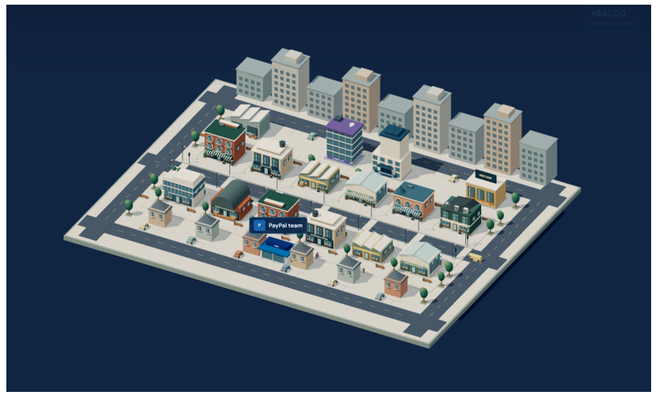
A bounded agent within a frozen commerce system

(a) Environment plane



s_t : shoppers, stock, cash and obligations

12 merchants; 600 simulated shoppers
Orders, deliveries and mature returns
Review barriers: displayed days 31 and 61

Allowed projection: o_t = observe(s_t)

o_t

policy a_t

(b) Agent plane: proposals, experiments and admission

measured feedback

Review state H_t
Prior trials and active contract

LLM agent
Select query, features, models, campaign

Typed call
`name + args`
`evidenceId`

bounded tool execution

Inspect demand
Signals, stock, contact funnel
Maturity-filtered outcomes

Fit model candidates
Two local model/feature recipes
Validation reports + modelHash

Evaluate campaign
3 separate development worlds
Hash-bound trial receipt

Deterministic admission gate
Typed fields + snapshot/model/trial identity + financial guards

publish best eligible candidate; otherwise retain

(c) Data and evidence plane

Warehouse + JSON artifact store

Records	Artifacts	State
orders / events / returns	trained models + reports	review checkpoints
merchant + customer IDs	trial receipts + hashes	model spend ledger

Observed evidence enters planning through bounded tools.

(d) Native PayPal Sandbox boundary

Vaulted wallet
Saved buyer consent

Orders / capture
Selected generated order

Verify + reconcile
IDs, amount, fee and net

order_id → capture_id → warehouse lineage

Native receipts are linked samples; simulation revenue is counted once.

SEALED CONFIRMATION 4 unseen seeds × {fixed rules, one agent, role-split team}

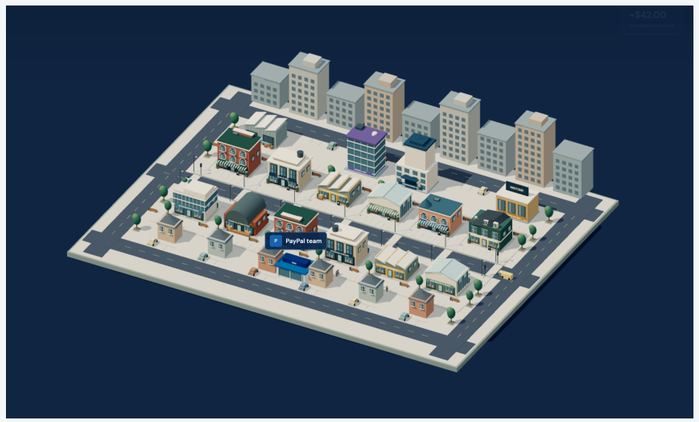
Measure settled PayPal contribution, merchant contribution, retained purchases and AI cost. Final scores do not feed the agent.

Figure 1. A hash-bound, checkpointed workflow separates LLM proposals from deterministic financial execution.

Symbols name existing V2.3 artifacts and state. The environment image is an actual recorded run; all market economics are simulated.

Market-to-agent loop with a controlled experiment sandbox

a Environment + observation plane



Observation projection

Signals • inventory • orders • mature outcomes
Read-only snapshot o_t

12 merchants / 600 shoppers / saved Day 32

observe

act

b Agent plane

trial feedback

Context

Snapshot + review history
Prior model / trial artifacts

LLM agent

Design an experiment
Propose or revise a campaign

Typed proposal

`campaign + modelHash`
`context.snapshotHash`

typed calls

LOCAL EXPERIMENT SANDBOX

1 Analyze

Inspect demand and funnel
Choose window + feature view

2 Train

Fit M_1 and M_2 locally
Compare validation evidence

3 Evaluate

Matched development trials
Return measured E_t

Historical data + mature in-run labels; no hidden shopper state or active-market future.

c Deterministic publication plane

Schema + scope **Artifact identity** **Financial guards**

Admit the best eligible tested contract, or keep the incumbent policy.

record outcomes

d Data plane

Warehouse + checkpoint store

Events / orders / returns • Models / reports / decision logs

Snapshot, model, contract and trial hashes preserve lineage.

e PayPal Sandbox + receipt lineage

Saved wallet ⇢ **Orders / capture** → **Verified receipt**

Recorded native example: $64.00 gross • $2.48 fee • $61.52 proceeds

Fictitious funds; receipt linked to an existing simulated order.

Figure 1. The LLM agent chooses experiments; trusted workers fit, test, validate and execute the resulting contracts.

Independent confirmation compares fixed, single-agent and role-split controls on four frozen unseen seeds; those final scores remain outside the loop.

Evidence-to-action protocol with typed checkpoints

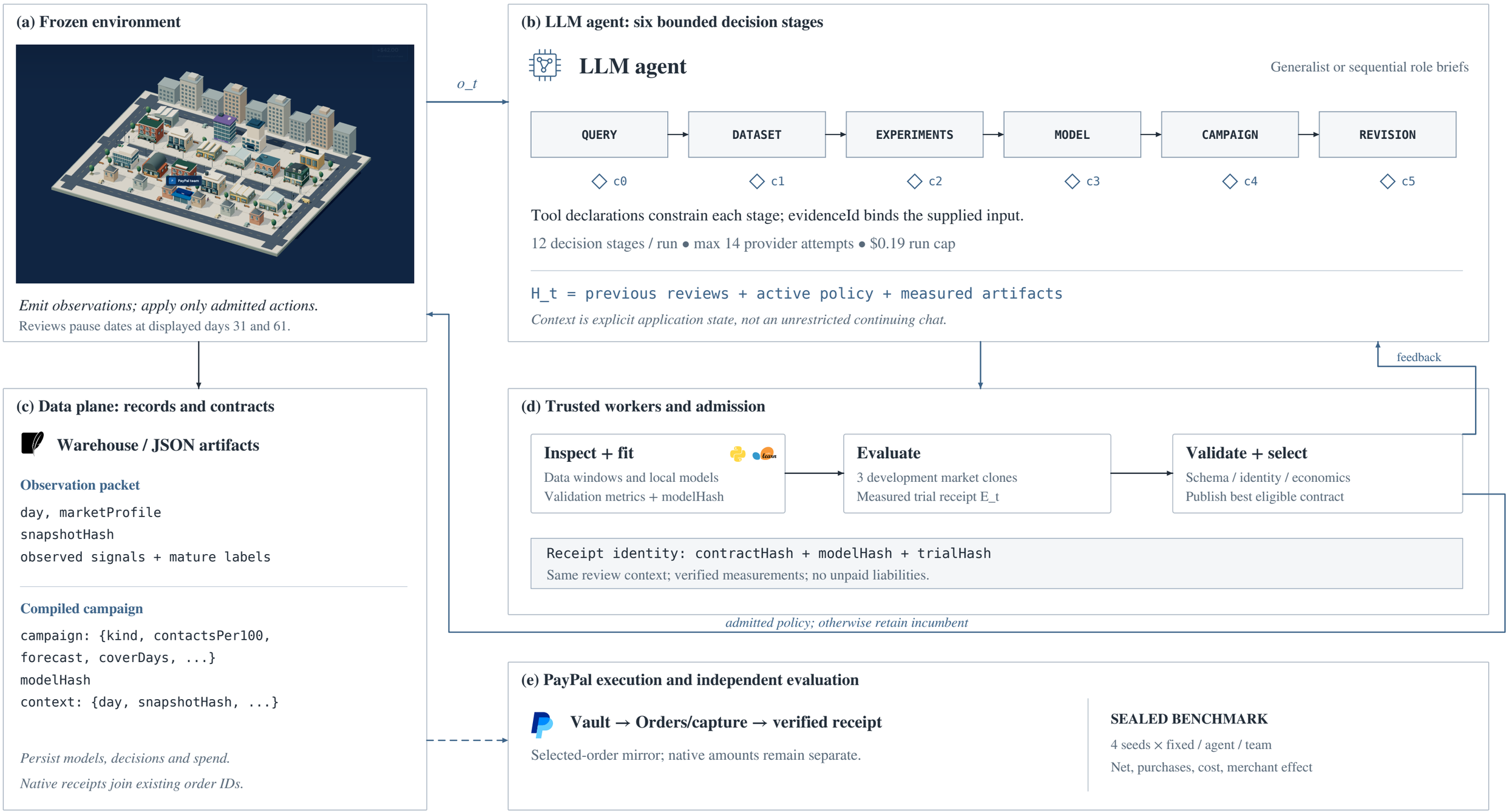


Figure 1. Typed contracts and checkpoints make model-driven choices replayable and constrain their execution.

Checkpoint markers denote recorded stages, not automatic crash-resume guarantees. The diagram uses current V2.3 interfaces and an actual saved environment.